ICT Class 7 - Formative Assessment 1

Formative Assessment (FA-1) - Answer Key

- Class 7

Class: Class 7

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Networking & Internet, Advanced Graphics

Curated and developed by

Hoysala T, Computer Science Teacher

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Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 marks)

Q.No	Answer	Explanation
1	b) LAN	LAN connects devices within a single building or small area. WAN covers large geographical areas.
2	c) SMTP	SMTP (Simple Mail Transfer Protocol) is used for sending emails.
3	a) Domain Name System	DNS translates domain names into IP addresses.
4	b) Fuzzy Select Tool	The Fuzzy Select Tool selects contiguous regions of similar color, useful for background removal.
5	d) All layers allow independent editing	In GIMP, each layer can be edited independently without affecting other layers.

Section II: Short Answer Questions (10 marks)

1. **What is the difference between LAN and WAN? Give one example of each.**
   - **Answer:** LAN (Local Area Network) connects devices within a small area like a school or office building. WAN (Wide Area Network) covers large geographical areas like cities or countries. Example of LAN: Network in a computer lab. Example of WAN: The Internet.
   - **Marking:** 1 mark for each correct definition with example (2 marks total).
2. **Name any four Internet services commonly used today.**
   - **Answer:** Email, World Wide Web (WWW), File Transfer Protocol (FTP), Instant Messaging, Video Conferencing, Online Banking.
   - **Marking:** 0.5 marks for each correct service (2 marks for four services).
3. **What is the purpose of a protocol in networking?**
   - **Answer:** Protocols are sets of rules that govern how data is transmitted between devices on a network. They ensure that all devices can communicate with each other in a standardized way, preventing errors and ensuring data integrity.
   - **Marking:** 1 mark for definition, 1 mark for explaining the purpose.
4. **Define the term “layer” in image editing. Why are layers useful?**
   - **Answer:** A layer is a transparent sheet on which different parts of an image are placed. Layers are useful because they allow editing one part of an image without affecting others, enable non-destructive editing, and make it easy to rearrange or combine elements.
   - **Marking:** 1 mark for definition, 1 mark for explaining usefulness.

5. **What is the difference between a raster image and a vector image?**
   - **Answer:** Raster images are made up of pixels and have fixed resolution (e.g., JPEG, PNG). Vector images are made up of mathematical paths and can be scaled without losing quality (e.g., SVG). Raster images are used for photographs, while vector images are used for logos and illustrations.
   - **Marking:** 1 mark for each type's description (2 marks total).

Section III: Long Answer Questions (5 marks)

1. **Explain the types of computer networks (LAN, MAN, WAN) with suitable examples and a diagram.**
   - **Answer:**
     - **LAN (Local Area Network):** Covers a small area like a building or campus. Examples: School computer lab, home network. Features: High speed, low cost, limited range.
     - **MAN (Metropolitan Area Network):** Covers a city or large campus. Examples: City-wide network connecting multiple schools. Features: Moderate speed, covers larger area than LAN.
     - **WAN (Wide Area Network):** Covers countries or continents. Examples: Internet, bank networks connecting branches. Features: Lower speed than LAN, expensive, unlimited range.
   - **Marking:** 2 marks for correct definitions, 2 marks for examples and features, 1 mark for diagram.

Part B: Hands-on Practical Lab Task (5 marks)

1. **Open GIMP and create a new image (800×600 pixels). Add two layers: one for background (fill with a color) and one for drawing a simple shape. Save the file as "LayerDemo.xcf".**
   - **Expected:** Image created with correct dimensions, two layers present, background filled with color, shape drawn on second layer, file saved in XCF format.
   - **Marking:** 1 mark for creating image with layers, 1 mark for correct save.
2. **Open a web browser and identify the protocol used when visiting "https://www.example.com". List the steps to find the IP address of a website using the command prompt.**
   - **Answer:** Protocol used is HTTPS (HyperText Transfer Protocol Secure). Steps: Open Command Prompt -> Type `nslookup www.example.com` -> Press Enter -> The IP address is displayed.
   - **Marking:** 1 mark for protocol identification, 1 mark for correct steps.
3. **Save your GIMP file on the desktop with the correct filename.**
   - **Expected:** File saved as "LayerDemo.xcf" on desktop.
   - **Marking:** 1 mark for correct filename and location.

Part C: Notebook Maintenance (5 marks)

Marks will be awarded for:

- Regularity in completing classwork (2 marks)
- Neatness of notes and diagrams (2 marks)
- Completion of all class activities (1 mark)

ICT Class 7 - Formative Assessment 2

Formative Assessment (FA-2) - Answer Key

- Class 7

Class: Class 7

Assessment Type: Formative Assessment (FA-2)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Multimedia, Data Handling

Curated and developed by

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Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 marks)

Q.No	Answer	Explanation
1	b) Audacity	Audacity is a free, open-source audio editing software.
2	a) MPEG-1 Audio Layer 3	MP3 stands for MPEG-1 Audio Layer 3, a common audio compression format.
3	c) =	In LibreOffice Calc, all formulas must start with an equals sign (=).
4	c) Line Chart	Line charts are best for showing trends and changes over time.
5	a) Adds all numbers in a range	The SUM function adds all numeric values in a specified range of cells.

Section II: Short Answer Questions (10 marks)

1. **Define multimedia. Name any three components of multimedia.**
   - **Answer:** Multimedia is the combination of different forms of media to present information. The five main components are: Text, Audio, Video, Graphics, and Animation.
   - **Marking:** 1 mark for definition, 0.33 marks for each component (1 mark for three).
2. **What is the difference between audio and video compression?**
   - **Answer:** Audio compression reduces the file size of audio data while maintaining quality (e.g., MP3). Video compression reduces the size of video files by removing redundant data frames (e.g., MP4). Both reduce storage space and bandwidth requirements.
   - **Marking:** 1 mark for audio compression explanation, 1 mark for video compression explanation.
3. **Write the syntax of the IF function in LibreOffice Calc with an example.**
   - **Answer:** Syntax: `=IF(condition; value_if_true; value_if_false)`. Example: `=IF(A1>=60; "Pass"; "Fail")` - If the value in A1 is 60 or above, it displays “Pass”, otherwise “Fail”.
   - **Marking:** 1 mark for correct syntax, 1 mark for practical example.
4. **What is a database? Name any two database management systems.**
   - **Answer:** A database is an organized collection of structured data stored electronically. Two DBMS: MySQL and Microsoft Access.
   - **Marking:** 1 mark for definition, 0.5 marks for each DBMS name.
5. **What is the purpose of animation in multimedia presentations?**

- **Answer:** Animation makes presentations more engaging and interactive. It helps explain concepts visually, captures audience attention, and makes complex information easier to understand through motion and visual effects.
- **Marking:** 2 marks for explaining purpose with examples.

Section III: Long Answer Questions (5 marks)

1. **Explain the steps to create a simple animation in Scratch. Describe the use of at least three different blocks in your animation.**
   - **Answer:**
     - Step 1: Open Scratch and choose a sprite.
     - Step 2: Add a "when green flag clicked" event block to start.
     - Step 3: Use "move 10 steps" block to move the sprite.
     - Step 4: Use "turn 15 degrees" block to change direction.
     - Step 5: Use "forever" loop block to repeat the animation continuously.
     - Step 6: Use "if on edge, bounce" to keep sprite within screen.
   - **Marking:** 1 mark for each step explained, 2 marks for describing three blocks.

Part B: Hands-on Practical Lab Task (5 marks)

1. **Open LibreOffice Calc and create a marks sheet for 5 students with columns: Name, Maths, Science, English, Total, Average. Use the SUM and AVERAGE formulas.**
   - **Expected:** Spreadsheet with 5 student entries, correct formulas for Total (SUM) and Average (AVERAGE).
   - **Marking:** 1 mark for correct structure, 1 mark for correct formulas.
2. **Open Audacity, record a 10-second voice clip, and apply the "Noise Reduction" effect. Export the file as "MyVoice.mp3".**
   - **Expected:** Audio recorded, Noise Reduction effect applied, file exported as MP3.
   - **Marking:** 1 mark for recording, 1 mark for effect application and export.
3. **Save your spreadsheet file on the desktop with the filename "MarksSheet.ods".**
   - **Expected:** File saved as "MarksSheet.ods" on desktop.
   - **Marking:** 1 mark for correct filename and location.

Part C: Notebook Maintenance (5 marks)

Marks will be awarded for:

- Regularity in completing classwork (2 marks)
- Neatness of notes and diagrams (2 marks)
- Completion of all class activities (1 mark)

ICT Class 7 - Formative Assessment 3

Formative Assessment (FA-3) - Answer Key

- Class 7

Class: Class 7

Assessment Type: Formative Assessment (FA-3)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Programming - Scratch, Digital Citizenship

Curated and developed by

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Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 marks)

Q.No	Answer	Explanation
1	a) create clone of myself	This block creates a duplicate copy of the sprite at runtime.
2	b) The trail of data left by online activities	A digital footprint is the record of all online activities a user engages in.
3	b) Hacking into someone's account	Hacking is unauthorized access to someone's computer or account, which is a cybercrime.
4	b) Runs the code repeatedly until stopped	The forever block continuously executes the blocks inside it until the program is stopped.
5	b) To protect the account from unauthorized access	Strong passwords prevent unauthorized users from accessing accounts.

Section II: Short Answer Questions (10 marks)

1. **What are “clones” in Scratch? How are they created?**
   - **Answer:** Clones are duplicate copies of a sprite created during runtime. They are created using the “create clone of myself” block. Clones can have their own scripts and behave independently from the original sprite.
   - **Marking:** 1 mark for definition, 1 mark for creation method.
2. **Name any two rules of cyber safety that every student should follow.**
   - **Answer:** 1) Never share personal information (name, address, phone number) online. 2) Never open emails or links from unknown senders. 3) Use strong passwords and change them regularly. 4) Always log out from accounts after use.
   - **Marking:** 0.5 marks for each correct rule (1 mark for two rules).
3. **What is the difference between a “local variable” and a “global variable” in Scratch?**
   - **Answer:** A local variable is accessible only within the sprite it is created for. A global variable (For all sprites) can be accessed and modified by any sprite in the project. Local variables are useful for sprite-specific data, while global variables are used for shared data like scores.
   - **Marking:** 1 mark for each variable type explanation.
4. **What is phishing? How can you identify a phishing email?**

- **Answer:** Phishing is a fraudulent attempt to obtain sensitive information like passwords and credit card details by pretending to be a trustworthy entity. Signs of phishing emails: urgent language, misspelled URLs, requests for personal information, unknown sender addresses, and suspicious attachments.
- **Marking:** 1 mark for definition, 1 mark for identification signs.

5. **Why is it important to create a custom block in Scratch? Give one advantage.**
   - **Answer:** Custom blocks allow you to create reusable code that can be called multiple times. One advantage is that it makes the code more organized and easier to debug. You can also pass parameters to make the block flexible for different situations.
   - **Marking:** 1 mark for importance, 1 mark for advantage.

Section III: Long Answer Questions (5 marks)

1. **Describe the steps to create a simple "Catch the Falling Objects" game in Scratch. Explain the role of at least four different blocks used in the game.**
   - **Answer:**
     - Step 1: Create a cat sprite as the catcher and a star sprite as the falling object.
     - Step 2: Use "when green flag clicked" event block to start the game.
     - Step 3: Use "forever" loop to make the star fall continuously.
     - Step 4: Use "change y by −5" block inside the loop to make the star move downward.
     - Step 5: Use "if key right arrow pressed" and "change x by 10" blocks to move the cat.
     - Step 6: Use "touching" block to detect collision and increase score.
     - Step 7: Use "if y position < −170" to detect when star reaches bottom and show "Game Over".
   - **Marking:** 1 mark for each step explained, 2 marks for explaining four blocks.

Part B: Hands-on Practical Lab Task (5 marks)

1. **Create a Scratch project where a cat sprite chases a mouse sprite using arrow keys. Use the "point towards" and "move" blocks.**
   - **Expected:** Cat sprite follows mouse, uses "point towards mouse" and "move steps" blocks, arrow key controls work.
   - **Marking:** 1 mark for sprite setup, 1 mark for correct blocks used.
2. **Create a Scratch project that uses at least 3 clones to display multiple stars on the screen. Use the "create clone of" block.**
   - **Expected:** At least 3 star clones visible on screen, "create clone of" block used correctly.
   - **Marking:** 1 mark for clone creation, 1 mark for correct implementation.
3. **Save your Scratch projects on the desktop with the filenames "ChaseGame.sb3" and "ClonesDemo.sb3".**
   - **Expected:** Both files saved on desktop with correct filenames.
   - **Marking:** 0.5 marks for each correct file.

Part C: Notebook Maintenance (5 marks)

Marks will be awarded for:

- Regularity in completing classwork (2 marks)
- Neatness of notes and diagrams (2 marks)
- Completion of all class activities (1 mark)

ICT Class 7 - Formative Assessment 4

Formative Assessment (FA-4) - Answer Key

- Class 7

Class: Class 7

Assessment Type: Formative Assessment (FA-4)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Programming - Scratch, Digital Citizenship, Revision

Curated and developed by

Hoysala T, Computer Science Teacher

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Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 marks)

Q.No	Answer	Explanation
1	a) play sound until done	This block plays a sound completely before moving to the next block.
2	b) Creative Commons	CC stands for Creative Commons, a licensing system for sharing creative works.
3	a) HyperText Transfer Protocol	HTTP is the protocol used for transferring web pages over the Internet.
4	c) Use a combination of letters, numbers, and symbols	Strong passwords contain a mix of characters to prevent unauthorized access.
5	d) XCF	XCF is GIMP's native format that preserves layers and other editing information.

Section II: Short Answer Questions (10 marks)

1. **What is the difference between "broadcast" and "broadcast and wait" blocks in Scratch?**
   - **Answer:** "Broadcast" sends a message to all sprites and continues executing immediately without waiting. "Broadcast and wait" sends a message and waits until all receiving sprites have finished their scripts before continuing. "Broadcast and wait" is useful when you need to ensure one action completes before the next begins.
   - **Marking:** 1 mark for each block explanation.
2. **Explain the concept of "digital footprint" and how it can affect your future.**
   - **Answer:** A digital footprint is the trail of data you leave behind when using the Internet, including social media posts, search history, and online purchases. It can affect your future because employers, colleges, and others may review your online activity. Negative content can harm your reputation and opportunities.
   - **Marking:** 1 mark for definition, 1 mark for explaining impact.
3. **What are the three types of network topologies? Name them.**
   - **Answer:** The three main network topologies are: 1) Bus Topology - all devices connected to a single cable. 2) Star Topology - all devices connected to a central hub or switch. 3) Ring Topology - devices connected in a circular pattern.
   - **Marking:** 0.67 marks for each correct topology name and description.

4. **Write any two uses of the AVERAGE function in LibreOffice Calc.**
   - **Answer:** 1) Calculating the average marks of students in a class. 2) Finding the average sales per month in a business report.
   - **Marking:** 1 mark for each correct use.
5. **What is the difference between “Save” and “Save As” in any software?**
   - **Answer:** “Save” overwrites the existing file with the current changes. “Save As” allows you to save the file with a new name, location, or format while keeping the original file unchanged.
   - **Marking:** 1 mark for “Save” explanation, 1 mark for “Save As” explanation.

Section III: Long Answer Questions (5 marks)

1. **Explain the difference between LAN, MAN, and WAN with suitable diagrams and examples. Which type of network is used in a school?**
   - **Answer:**
     - **LAN (Local Area Network):** Covers a small area like a building. High speed, low cost. Example: Computer lab in school.
     - **MAN (Metropolitan Area Network):** Covers a city. Moderate speed. Example: Network connecting branches of a bank in a city.
     - **WAN (Wide Area Network):** Covers countries or continents. Lower speed, high cost. Example: The Internet.
     - A school typically uses a LAN to connect computers within the campus.
   - **Marking:** 1 mark for each topology definition, 1 mark for examples, 1 mark for diagram and school example.

Part B: Hands-on Practical Lab Task (5 marks)

1. **Create a Scratch project that uses a custom block to draw a square shape using the pen extension.**
   - **Expected:** Custom block defined, pen extension used, square drawn with equal sides and 90-degree turns.
   - **Marking:** 1 mark for custom block, 1 mark for correct square drawing.
2. **Open a web browser, search for information on “cyber safety tips for students”, and write down five tips in a text file.**
   - **Expected:** Five relevant cyber safety tips documented in a text file.
   - **Marking:** 1 mark for research, 1 mark for five correct tips.
3. **Save both files on the desktop with appropriate filenames.**
   - **Expected:** Both files saved on desktop with meaningful filenames.
   - **Marking:** 1 mark for correct file saving.

Part C: Notebook Maintenance (5 marks)

Marks will be awarded for:

- Regularity in completing classwork (2 marks)

- Neatness of notes and diagrams (2 marks)
- Completion of all class activities (1 mark)

ICT Class 7 - Practical Lab Exam 1

Practical Lab Exam 1 - Answer Key

- Class 7

Class: Class 7

Assessment Type: Practical Lab Exam 1

Total Marks: 20

Duration: 45 minutes

Topics Covered: Networking & Internet, Advanced Graphics, Multimedia, Data Handling

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Answer Key - Grading Guidelines

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Lab Tasks (15 marks)

Task 1: Network Identification (3 marks)

1. **Expected Steps:**
   - Open Command Prompt.
   - Type `ipconfig` and press Enter.
   - Note IPv4 Address and Default Gateway.
2. **Expected Answers:**
   - IPv4 Address: [Specific to computer, e.g., 192.168.1.100]
   - Default Gateway: [Specific to network, e.g., 192.168.1.1]
   - Network type: LAN (if within same building) or WAN (if connected to Internet)
3. **Marking:**
   - 1 mark for opening Command Prompt and running command.
   - 1 mark for noting IPv4 Address.
   - 1 mark for identifying network type.

Task 2: Image Editing in GIMP (4 marks)

1. **Expected Steps:**
   - Open GIMP -> File -> New -> Set dimensions 600×400 pixels.
   - Create three layers: Background, Shape, Text.
   - Background layer: Select Blend Tool -> Apply gradient.
   - Shape layer: Use Rectangle Select Tool -> Draw rectangle -> Use Ellipse Select Tool -> Draw circle.
   - Text layer: Select Text Tool -> Click on canvas -> Type name.
   - File -> Save As -> “MyArt.xcf” -> Save to Desktop.
2. **Marking:**
   - 1 mark for creating image with correct dimensions.
   - 1 mark for creating three layers.
   - 1 mark for drawing shapes and adding text.
   - 1 mark for saving as XCF format.

Task 3: Audio Recording in Audacity (3 marks)

1. **Expected Steps:**
   - Open Audacity -> Click Record button -> Record 15 seconds -> Click Stop.
   - Select audio -> Effect -> Noise Reduction -> Get Noise Profile -> Apply.

- Select audio -> Effect -> Amplify -> Adjust and apply.
- File -> Export -> Export as MP3 -> “MyVoice.mp3” -> Save to Desktop.

2. **Marking:**
   - 1 mark for recording audio.
   - 1 mark for applying Noise Reduction and Amplify effects.
   - 1 mark for exporting as MP3.

Task 4: Spreadsheet in LibreOffice Calc (5 marks)

1. **Expected Table Structure:**

   | Student Name | Maths | Science | English | Total | Average | |————-|——-|———|———|——-|———| | Rahul | 85 | 78 | 90 | 253 | 84.33 | | Priya | 92 | 88 | 85 | 265 | 88.33 | | Arjun | 78 | 92 | 88 | 258 | 86.00 |

2. **Expected Formulas:**
   - Total: `=SUM(B2:D2)` for Rahul (and similar for others)
   - Average: `=AVERAGE(B2:D2)` for Rahul (and similar for others)
3. **Expected Chart:**
   - Bar chart comparing Maths, Science, English marks for all three students.
4. **Marking:**
   - 1 mark for correct table structure.
   - 1 mark for SUM formula.
   - 1 mark for AVERAGE formula.
   - 1 mark for bar chart creation.
   - 1 mark for saving as “MarksAnalysis.ods”.

Viva Voce (5 marks)

1. **What is the difference between LAN and WAN?**
   - **Answer:** LAN covers a small area (building), WAN covers large areas (cities/countries). (1 mark)
2. **Why are layers important in image editing?**
   - **Answer:** Layers allow non-destructive editing, editing parts independently, and easy rearrangement. (1 mark)
3. **What is the use of the SUM function in a spreadsheet?**
   - **Answer:** SUM adds all numeric values in a specified range of cells. (1 mark)
4. **Name any two audio formats.**
   - **Answer:** MP3, WAV, AAC, FLAC. (1 mark)
5. **What is the purpose of a firewall in a network?**
   - **Answer:** A firewall monitors and controls network traffic to protect against unauthorized access. (1 mark)

ICT Class 7 - Practical Lab Exam 2

Practical Lab Exam 2 - Answer Key

- Class 7

Class: Class 7

Assessment Type: Practical Lab Exam 2

Total Marks: 20

Duration: 45 minutes

Topics Covered: Complete Syllabus - All Practical Activities

Curated and developed by

Hoysala T, Computer Science Teacher

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Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Lab Tasks (15 marks)

Task 1: Scratch Game Development (5 marks)

1. **Cat sprite movement:**
   - When right arrow key pressed -> change x by 10.
   - When left arrow key pressed -> change x by −10.
2. **Star falling:**
   - When green flag clicked -> go to random position at top -> forever -> change y by −5 -> if y position < −170 -> go to random position at top.
3. **Score increase:**
   - When cat touches star -> change score by 1 -> go to random position at top.
4. **Game Over:**
   - When star touches edge -> say “Game Over” for 2 seconds -> stop all.
5. **Save:**
   - File -> Save to Desktop as “CatchStars.sb3”.
6. **Marking:**
   - 1 mark for cat movement with arrow keys.
   - 1 mark for star falling randomly.
   - 1 mark for score increment on catch.
   - 1 mark for Game Over display.
   - 1 mark for correct save.

Task 2: Advanced GIMP Editing (4 marks)

1. **Expected Steps:**
   - Open image -> Image -> Duplicate -> Work on copy.
   - Select background layer -> Filters -> Blur -> Gaussian Blur -> Apply.
   - Use Fuzzy Select Tool -> Click on background -> Edit -> Clear (remove background).
   - Select Text Tool -> Click on image -> Type name -> Choose different color.
   - File -> Save As -> “EditedPhoto.xcf” -> Save to Desktop.
2. **Marking:**
   - 1 mark for duplicating image.
   - 1 mark for applying Gaussian Blur.
   - 1 mark for removing background with Fuzzy Select.
   - 1 mark for adding text overlay and saving.

Task 3: Data Analysis in LibreOffice Calc (4 marks)

1. **Expected Table Structure:**

   | Roll No | Name | Class | Marks | Grade | Status | |———|———|——-|——-|——-|———| | 1 | Rahul | 7A | 85 | B | Pass | | 2 | Priya | 7A | 92 | A | Pass | | 3 | Arjun | 7B | 58 | D | Pass | | 4 | Meera | 7B | 72 | C | Pass | | 5 | Kiran | 7A | 30 | D | Fail |

2. **Expected Formulas:**
   - Grade: `=IF(E2>=90; "A"; IF(E2>=75; "B"; IF(E2>=60; "C"; "D")))`
   - Status: `=IF(E2>=35; "Pass"; "Fail")`
3. **Expected Chart:**
   - Pie chart showing grade distribution (A, B, C, D counts).
4. **Marking:**
   - 1 mark for correct table structure with 5 students.
   - 1 mark for Grade IF formula.
   - 1 mark for Status IF formula.
   - 1 mark for pie chart and save.

Task 4: Internet Research and Documentation (2 marks)

1. **Expected Content:**
   - Title: "Online Safety Guide"
   - Five safety tips:
     1. Never share personal information online.
     2. Use strong passwords for all accounts.
     3. Do not click on unknown links or attachments.
     4. Always log out after using public computers.
     5. Report any suspicious activity to a trusted adult.
2. **Marking:**
   - 1 mark for research and document creation.
   - 1 mark for five relevant safety tips with explanations.

Viva Voce (5 marks)

1. **What are clones in Scratch? How are they created?**
   - **Answer:** Clones are duplicate sprites created during runtime using the "create clone of myself" block. (1 mark)
2. **Explain the difference between "Save" and "Save As" in any software.**
   - **Answer:** Save overwrites the current file. Save As creates a new file with a different name or location. (1 mark)
3. **What is the IF function? Give a practical example of its use.**
   - **Answer:** IF checks a condition and returns different values based on the result. Example: `=IF(A1>=60; "Pass"; "Fail")`. (1 mark)
4. **What are the ethical considerations when using images from the Internet?**

- **Answer:** Check copyright status, give credit to the creator, use Creative Commons licensed images, and do not use images without permission. (1 mark)

5. **Explain the concept of "digital footprint" with a real-life example.**
   - **Answer:** Digital footprint is the trail of data left by online activities. Example: Social media posts, search history, or online purchases that can be traced back to you. (1 mark)

ICT Class 7 - Summative Assessment 1

Summative Assessment (SA-1) - Answer Key

- Class 7

Class: Class 7

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Ch 1: Networking & Internet, Ch 2: Advanced Graphics, Ch 3: Multimedia, Ch 4: Data Handling

Curated and developed by

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Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Multiple Choice Questions (10 marks)

Q.No	Answer	Explanation
1	c) MAN	MAN (Metropolitan Area Network) spans across a city.
2	b) HTTPS	HTTPS (HyperText Transfer Protocol Secure) provides secure web browsing with encryption.
3	a) Uniform Resource Locator	URL stands for Uniform Resource Locator, the address of a web page.
4	b) Rectangle Select Tool	The Rectangle Select Tool is used to select rectangular areas in GIMP.
5	c) .mp3	.mp3 is the most common audio format for music files.
6	a) Add values from A1 to A5	=SUM(A1:A5) adds all numeric values in cells A1 through A5.
7	b) Inkscape	Inkscape is a vector graphics editor. GIMP is raster, Audacity is audio, VLC is media player.
8	b) To protect a network from unauthorized access	A firewall monitors and controls incoming and outgoing network traffic.
9	b) Automatically calculates the sum of selected cells	AutoSum quickly adds selected cells using the SUM function.
10	c) Video	Video deals with moving pictures combined with sound.

Part B: Short Answer Questions (16 marks)

1. **Differentiate between LAN and WAN with one example each.**
   - **Answer:** LAN connects devices within a small area like a building. WAN covers large geographical areas like countries. Example of LAN: School computer lab. Example of WAN: Internet connecting computers worldwide.
   - **Marking:** 1 mark for LAN, 1 mark for WAN with examples.
2. **What is a search engine? Name any two search engines.**
   - **Answer:** A search engine is a software system that searches the Internet for information based on user queries. Two search engines: Google, Bing.
   - **Marking:** 1 mark for definition, 1 mark for names.

3. **Explain the use of layers in GIMP. Why are they important for image editing?**
   - **Answer:** Layers are transparent sheets on which different parts of an image are placed. They are important because they allow non-destructive editing, enable editing one part without affecting others, and make it easy to rearrange elements.
   - **Marking:** 1 mark for definition, 1 mark for importance.
4. **What is the difference between raster and vector graphics? Give one example of each.**
   - **Answer:** Raster graphics are made of pixels and have fixed resolution (e.g., JPEG photos). Vector graphics use mathematical paths and scale without quality loss (e.g., SVG logos).
   - **Marking:** 1 mark for each type with example.
5. **Define multimedia. List its five main components.**
   - **Answer:** Multimedia is the integration of multiple media forms. Components: Text, Audio, Video, Graphics, Animation.
   - **Marking:** 1 mark for definition, 1 mark for components.
6. **Write the syntax of the IF function in LibreOffice Calc with a practical example.**
   - **Answer:** Syntax: `=IF(condition; value_if_true; value_if_false)`. Example: `=IF(A1>=60; "Pass"; "Fail")`.
   - **Marking:** 1 mark for syntax, 1 mark for example.
7. **What is the difference between a formula and a function in a spreadsheet?**
   - **Answer:** A formula is a user-defined expression using operators (e.g., =A1+B1). A function is a predefined formula provided by the software (e.g., SUM, AVERAGE).
   - **Marking:** 1 mark for formula, 1 mark for function.
8. **What are the advantages of using charts in data analysis?**
   - **Answer:** Charts help visualize data trends, make comparisons easier, identify patterns, present data clearly to audiences, and simplify complex data sets.
   - **Marking:** 0.4 marks for each advantage (2 marks for five advantages).

Part C: Long Answer Questions (12 marks)

1. **Explain the types of computer networks (LAN, MAN, WAN) with a neat diagram. Compare their features, advantages, and disadvantages.**
   - **Answer:**
     - **LAN:** Small area, high speed, low cost. Advantages: Resource sharing, easy maintenance. Disadvantages: Limited range, security risks if not configured.
     - **MAN:** City-wide, moderate speed, higher cost. Advantages: Connects multiple LANs, covers larger area. Disadvantages: More complex setup, higher maintenance.
     - **WAN:** Global, lower speed, highest cost. Advantages: Global connectivity, unlimited range. Disadvantages: Expensive, slower speeds, security concerns.
   - **Marking:** 1 mark for each definition, 1 mark for features comparison, 1 mark for diagram.
2. **Describe the steps to edit an image in GIMP using layers. Explain how to add a new layer, draw on it, and merge layers.**
   - **Answer:**
     - Step 1: Open image in GIMP.

    - Step 2: Go to Layer menu -> New Layer to add a new layer.
    - Step 3: Select the new layer and use drawing tools to add content.
    - Step 4: Use Layer menu -> Merge Down to combine layers.
    - Step 5: Save the edited image.
  - **Marking:** 2 marks for steps, 1 mark for explaining merge process.
3. **What is multimedia? Explain the role of audio and video in multimedia with examples. How is compression used in multimedia?**
  - **Answer:**
    - Multimedia combines text, audio, video, graphics, and animation.
    - **Audio role:** Adds sound effects, narration, and music (e.g., background music in presentations).
    - **Video role:** Provides visual storytelling and demonstrations (e.g., educational videos).
    - **Compression:** Reduces file size for storage and transmission (e.g., MP3 for audio, MP4 for video).
  - **Marking:** 1 mark for definition, 1 mark for audio and video roles, 1 mark for compression.
4. **Explain the components of a spreadsheet with a suitable example. Describe how to use SUM, AVERAGE, MAX, and MIN functions.**
  - **Answer:**
    - Components: Cells, rows, columns, sheets, formulas, functions.
    - **SUM:** Adds values (e.g., =SUM(A1:A10)).
    - **AVERAGE:** Finds mean value (e.g., =AVERAGE(B1:B10)).
    - **MAX:** Returns largest value (e.g., =MAX(C1:C10)).
    - **MIN:** Returns smallest value (e.g., =MIN(D1:D10)).
  - **Marking:** 1 mark for components, 0.5 marks for each function.

Part D: Application Based Questions (2 marks)

1. **Raj wants to create a photo album with multiple images. Which image editing software would you recommend and why? Mention two features that would help him.**
  - **Answer:** Recommend GIMP because it is free and supports layers. Features: 1) Layer support for organizing multiple images. 2) Filters and effects to enhance photos.
  - **Marking:** 1 mark for recommendation, 1 mark for features.
2. **A school wants to connect computers in 5 different buildings across a campus. Which type of network would be most suitable? Justify your answer.**
  - **Answer:** MAN (Metropolitan Area Network) is most suitable because it covers a campus area larger than a single building but smaller than a city. It provides high-speed connectivity between buildings.
  - **Marking:** 1 mark for correct network type, 1 mark for justification.

ICT Class 7 - Summative Assessment 2

Summative Assessment (SA-2) - Answer Key

- Class 7

Class: Class 7

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Ch 1-6: Complete Syllabus

Curated and developed by

Hoysala T, Computer Science Teacher

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Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Multiple Choice Questions (10 marks)

Q.No	Answer	Explanation
1	c) Sound	The Sound category contains blocks for playing sounds and controlling volume.
2	b) Creative Commons	CC stands for Creative Commons, a licensing framework for sharing creative works.
3	b) POP3	POP3 (Post Office Protocol 3) is used to receive emails from a mail server.
4	b) Gaussian Blur	Gaussian Blur is a filter that creates a smooth blur effect on images.
5	a) Find the total of a range of cells	The SUM function adds all numeric values in a specified range.
6	b) The trail of data left by a user's online activities	A digital footprint is the record of all online activities.
7	a) create clone of myself	This block creates a duplicate copy of the current sprite.
8	d) Tree Topology	Star, Bus, and Ring are common topologies. Tree is a hierarchical structure, not a basic topology type.
9	b) Stops all scripts in the project	The stop all block halts every running script in the entire project.
10	c) Keeping software and antivirus updated	Updating software and antivirus is a safe online practice that protects against threats.

Part B: Short Answer Questions (16 marks)

1. **What is the difference between a client and a server in a network?**
   - **Answer:** A client is a computer that requests services (e.g., browsing a website). A server is a computer that provides services to clients (e.g., hosting a website).
   - **Marking:** 1 mark for client, 1 mark for server.
2. **Explain any two Internet services with their uses.**
   - **Answer:** 1) Email - for sending and receiving messages electronically. 2) FTP - for transferring files between computers over the Internet.
   - **Marking:** 1 mark for each service with use.

3. **What are layers in GIMP? How do they help in editing images?**
   - **Answer:** Layers are transparent sheets for organizing image elements. They help by allowing non-destructive editing, editing individual parts without affecting others, and easy rearrangement.
   - **Marking:** 1 mark for definition, 1 mark for explaining benefits.
4. **What is the difference between JPEG and PNG image formats?**
   - **Answer:** JPEG uses lossy compression, smaller file size, good for photographs. PNG uses lossless compression, supports transparency, good for graphics and logos.
   - **Marking:** 1 mark for JPEG, 1 mark for PNG.
5. **Explain the concept of animation. What are the two main types of animation?**
   - **Answer:** Animation creates the illusion of movement using sequences of images. Two types: 1) Frame animation - individual images displayed in sequence. 2) Vector animation - mathematical paths define movement.
   - **Marking:** 1 mark for concept, 1 mark for types.
6. **What is a formula in a spreadsheet? Write the syntax of the AVERAGE function.**
   - **Answer:** A formula is a calculation expression in a spreadsheet. Syntax: `=AVERAGE(range)`. Example: `=AVERAGE(A1:A10)`.
   - **Marking:** 1 mark for definition, 1 mark for syntax.
7. **What are "clones" in Scratch? How are they useful in game development?**
   - **Answer:** Clones are duplicate sprites created during runtime. They are useful for creating multiple similar objects (e.g., bullets, enemies, stars) without manually creating each one.
   - **Marking:** 1 mark for definition, 1 mark for usefulness.
8. **What is plagiarism? Why should students avoid it?**
   - **Answer:** Plagiarism is using someone's work without credit. Students should avoid it because it is dishonest, can result in academic penalties, and violates intellectual property rights.
   - **Marking:** 1 mark for definition, 1 mark for reasons to avoid.

Part C: Long Answer Questions (12 marks)

1. **Explain the complete anatomy of a computer network with a diagram. Include the role of routers, switches, and modems in a network.**
   - **Answer:**
     - **Routers:** Direct data packets between different networks.
     - **Switches:** Connect devices within a LAN and manage data flow.
     - **Modems:** Convert digital signals to analog for transmission over phone lines.
     - Diagram should show these components connected with proper labels.
   - **Marking:** 2 marks for component explanations, 1 mark for diagram.
2. **Describe the steps to create an animated story in Scratch. Include the use of at least four different block categories.**
   - **Answer:**
     - Step 1: Choose sprites and backgrounds.

    - Step 2: Use Events blocks to start the story.
    - Step 3: Use Looks blocks for speech bubbles and costumes.
    - Step 4: Use Motion blocks for sprite movement.
    - Step 5: Use Control blocks for timing and loops.
    - Step 6: Use Sound blocks for background music.
  - **Marking:** 2 marks for steps, 1 mark for block categories.
3. **What is data handling? Explain the steps to create a dashboard in LibreOffice Calc using charts and formulas. Give a practical example.**
  - **Answer:**
    - Data handling involves collecting, organizing, analyzing, and presenting data.
    - Steps: 1) Enter data in cells. 2) Use formulas (SUM, AVERAGE) for calculations. 3) Select data range. 4) Insert chart (bar, pie, line). 5) Customize chart labels and colors.
    - Example: Student marks analysis with total, average, and grade charts.
  - **Marking:** 1 mark for data handling definition, 2 marks for steps and example.
4. **Explain the importance of digital citizenship. Describe any five rules of responsible technology use with examples.**
  - **Answer:**
    - Digital citizenship means using technology responsibly and ethically.
    - Rules: 1) Protect personal information online. 2) Think before you post. 3) Respect others online. 4) Do not plagiarize. 5) Report cyberbullying.
  - **Marking:** 1 mark for importance, 0.4 marks for each rule (2 marks for five rules).

Part D: Application Based Questions (2 marks)

1. **Meera wants to create a presentation with background music and animated images. Which multimedia tools would you recommend? Explain how she can use them.**
  - **Answer:** Recommend: 1) LibreOffice Impress for presentation creation. 2) Audacity for editing background music. 3) GIMP for editing animated images. She can add audio files to slides and create animated GIFs.
  - **Marking:** 1 mark for tool recommendations, 1 mark for usage explanation.
2. **A student receives an email asking for their bank details with a link to “update account information”. What should the student do and why?**
  - **Answer:** The student should NOT click the link or provide any information. This is a phishing attempt. The student should report the email to a teacher or parent and delete it immediately.
  - **Marking:** 1 mark for correct action, 1 mark for explaining why.

ICT Class 7 - Unit Test 1

Unit Test 1 - Answer Key

- Class 7

Class: Class 7

Assessment Type: Unit Test 1

Total Marks: 10

Duration: 30 minutes

Topics Covered: Networking & Internet

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Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: Multiple Choice Questions (5 marks)

Q.No	Answer	Explanation
1	b) LAN	LAN (Local Area Network) covers a small geographical area like a single building.
2	c) MS Word	MS Word is a word processing software, not an Internet service.
3	a) Internet Service Provider	ISP stands for Internet Service Provider, the company that provides Internet access.
4	c) FTP	FTP (File Transfer Protocol) is used to transfer files between computers over a network.
5	a) Metropolitan Area Network	MAN stands for Metropolitan Area Network, covering a city or large campus.

Section II: Short Answer Questions (5 marks)

1. **What is a computer network? State its two advantages.**
   - **Answer:** A computer network is a collection of interconnected computers that share resources and data. Two advantages: 1) Resource sharing - multiple users can share printers and files. 2) Communication - enables email and instant messaging between users.
   - **Marking:** 1 mark for definition, 0.5 marks for each advantage.
2. **Differentiate between client and server in a network.**
   - **Answer:** A client is a computer that requests services or resources from another computer. A server is a computer that provides services or resources to clients. Example: When you browse a website, your computer is the client and the website's computer is the server.
   - **Marking:** 1 mark for client definition, 1 mark for server definition.
3. **What is the role of a modem in a computer network?**
   - **Answer:** A modem (Modulator-Demodulator) converts digital signals from a computer into analog signals for transmission over telephone lines, and vice versa. It enables computers to connect to the Internet through phone lines or cable connections.
   - **Marking:** 1 mark for explaining conversion, 1 mark for stating its purpose.
4. **Name any two web browsers and one search engine.**
   - **Answer:** Web browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari. Search engine: Google, Bing, Yahoo.
   - **Marking:** 0.5 marks for each browser, 0.5 marks for search engine.

5. **What is an IP address? Why is it important?**
    - **Answer:** An IP (Internet Protocol) address is a unique numerical identifier assigned to each device on a network. It is important because it allows devices to identify and communicate with each other over the Internet, similar to a postal address for mail delivery.
    - **Marking:** 1 mark for definition, 1 mark for explaining importance.

ICT Class 7 - Unit Test 2

Unit Test 2 - Answer Key

- Class 7

Class: Class 7

Assessment Type: Unit Test 2

Total Marks: 10

Duration: 30 minutes

Topics Covered: Multimedia

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Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: Multiple Choice Questions (5 marks)

Q.No	Answer	Explanation
1	b) Audacity	Audacity is a free, open-source audio editing software.
2	a) Musical Instrument Digital Interface	MIDI stands for Musical Instrument Digital Interface, a standard for connecting electronic musical instruments.
3	a) .mp4	.mp4 is the file extension for MP4 video files.
4	c) Video	Video involves moving images combined with sound to create multimedia content.
5	b) Reduce file size	Compression reduces file size to save storage space and enable faster transmission.

Section II: Short Answer Questions (5 marks)

1. **What is multimedia? List its five components.**
    - **Answer:** Multimedia is the integration of multiple forms of media to present information in an engaging way. Five components: Text, Audio, Video, Graphics, and Animation.
    - **Marking:** 1 mark for definition, 0.2 marks for each component (1 mark for five).
2. **What is the difference between WAV and MP3 audio formats?**
    - **Answer:** WAV is an uncompressed audio format with high quality but large file size. MP3 is a compressed audio format with slightly lower quality but much smaller file size. WAV is used for professional recording, while MP3 is used for everyday listening.
    - **Marking:** 1 mark for WAV description, 1 mark for MP3 description.
3. **Define animation. Give one real-life example of animation.**
    - **Answer:** Animation is the technique of creating the illusion of movement by displaying a sequence of images or frames in rapid succession. Example: Animated cartoons like Chhota Bheem, or traffic lights changing colors.
    - **Marking:** 1 mark for definition, 1 mark for example.
4. **What is the purpose of a media player?**
    - **Answer:** A media player is software used to play audio and video files. It allows users to listen to music, watch videos, and view photos. Examples include VLC Media Player and Windows Media Player.
    - **Marking:** 1 mark for purpose, 1 mark for examples.
5. **What are the uses of multimedia in education?**

- **Answer:** Multimedia in education makes learning interactive and engaging through: 1) Educational videos and animations. 2) Interactive simulations and experiments. 3) Audio-visual presentations. 4) E-books with embedded media. 5) Online courses and virtual labs.
- **Marking:** 0.4 marks for each use (2 marks for five uses).

ICT Class 7 - Unit Test 3

Unit Test 3 - Answer Key

- Class 7

Class: Class 7

Assessment Type: Unit Test 3

Total Marks: 10

Duration: 30 minutes

Topics Covered: Programming - Scratch

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Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: Multiple Choice Questions (5 marks)

Q.No	Answer	Explanation
1	b) Looks	The Looks category contains blocks for displaying speech bubbles, thinking, and changing costumes.
2	b) Cat	The default sprite in a new Scratch project is the Cat sprite.
3	b) repeat 10	The repeat block executes a set of instructions a specific number of times.
4	a) create clone of [myself]	This block creates a duplicate copy of the current sprite.
5	b) when green flag clicked	This event block starts the program when the green flag is clicked.

Section II: Short Answer Questions (5 marks)

1. **What is Scratch? Who developed it?**
    - **Answer:** Scratch is a free, visual programming language developed by MIT Media Lab. It allows users to create interactive stories, games, and animations using drag-and-drop code blocks.
    - **Marking:** 1 mark for definition, 1 mark for developer and features.
2. **Explain the difference between “Motion” and “Looks” blocks with one example each.**
    - **Answer:** Motion blocks control the movement of sprites (e.g., “move 10 steps”). Looks blocks control the appearance of sprites (e.g., “say Hello for 2 seconds”).
    - **Marking:** 1 mark for Motion explanation with example, 1 mark for Looks explanation with example.
3. **What is a “loop” in programming? Name two types of loops available in Scratch.**
    - **Answer:** A loop is a control structure that repeats a set of instructions. Two types in Scratch: 1) Forever loop - repeats indefinitely until stopped. 2) Repeat loop - repeats a specific number of times.
    - **Marking:** 1 mark for definition, 0.5 marks for each loop type.
4. **How do you create a custom block in Scratch? Write the steps.**
    - **Answer:** Steps: 1) Go to “My Blocks” category. 2) Click “Make a Block”. 3) Enter a name for the block. 4) Add parameters if needed. 5) Click OK. 6) Define the block’s behavior using other blocks.
    - **Marking:** 2 marks for correct steps.
5. **What is the purpose of the “pen” extension in Scratch?**

- **Answer:** The pen extension allows sprites to draw on the stage as they move. It can be used to create patterns, draw shapes, and create art. The pen can be lifted, put down, and its color and size can be changed.
- **Marking:** 2 marks for explaining purpose and features.

ICT Class 7 - Unit Test 4

Unit Test 4 - Answer Key

- Class 7

Class: Class 7

Assessment Type: Unit Test 4

Total Marks: 10

Duration: 30 minutes

Topics Covered: Digital Citizenship

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Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: Multiple Choice Questions (5 marks)

Q.No	Answer	Explanation
1	b) Posting hurtful comments about someone online	Cyberbullying involves using digital devices to harass or harm others.
2	c) A mix of uppercase, lowercase, numbers, and symbols	Strong passwords contain a variety of character types for security.
3	b) A fraudulent attempt to obtain sensitive information	Phishing is a cyber attack that tricks users into revealing personal data.
4	c) Both (a) and (b)	Both firewalls and antivirus software protect computers from threats.
5	b) The creator has exclusive rights over their original work	Copyright gives creators legal protection over their creative works.

Section II: Short Answer Questions (5 marks)

1. **What is cyber safety? Why is it important for students?**
   - **Answer:** Cyber safety refers to the practices and measures taken to protect yourself and your information while using the Internet and digital devices. It is important for students because they spend significant time online, and without proper knowledge, they may become victims of cyberbullying, phishing, or identity theft.
   - **Marking:** 1 mark for definition, 1 mark for explaining importance.
2. **Explain the term “digital footprint” with an example.**
   - **Answer:** A digital footprint is the trail of data you leave behind when using the Internet. Example: When you post a photo on social media, comment on a website, or search for information on Google, you leave a digital footprint that can be tracked and may remain permanently.
   - **Marking:** 1 mark for definition, 1 mark for example.
3. **What are the rules of netiquette? State any three.**
   - **Answer:** Netiquette rules: 1) Be respectful in online communications. 2) Do not use ALL CAPS (considered shouting). 3) Think before you post - content can be permanent. 4) Do not share others’ private information. 5) Use appropriate language in online interactions.
   - **Marking:** 0.67 marks for each correct rule (2 marks for three rules).
4. **What is plagiarism? How can it be avoided?**

- **Answer:** Plagiarism is using someone else's work or ideas without giving them proper credit. It can be avoided by: 1) Always citing sources. 2) Using quotation marks for direct quotes. 3) Writing content in your own words. 4) Using plagiarism detection tools.
- **Marking:** 1 mark for definition, 1 mark for avoidance methods.

5. **Name any two ways to protect personal information online.**
   - **Answer:** 1) Use strong, unique passwords for different accounts. 2) Enable two-factor authentication. 3) Do not share personal details on social media. 4) Use privacy settings on social media platforms.
   - **Marking:** 0.5 marks for each correct way (1 mark for two ways).